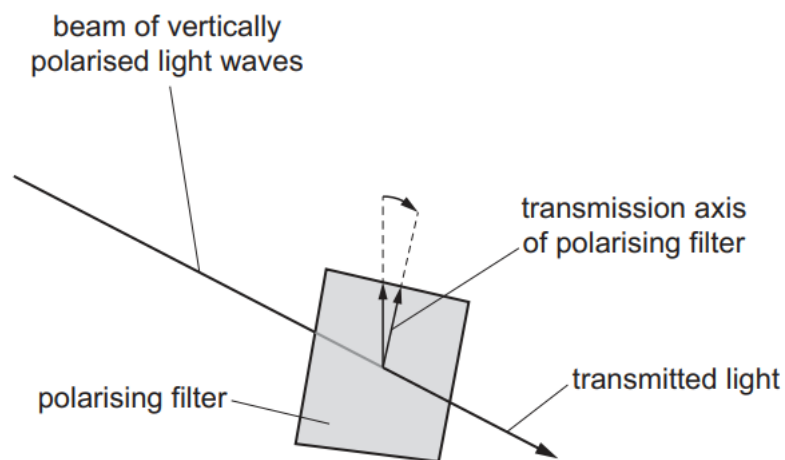


- 19 A beam of vertically polarised light is incident normally on a polarising filter. The filter can be rotated so that it is always in a plane perpendicular to the beam. The transmission axis of the filter is initially vertical.



The filter is first rotated clockwise by an angle of 30° so that the transmitted light waves have intensity I_{30} . The filter is then rotated clockwise by a further angle of 30° .

What is the new intensity of the transmitted light waves?

- A $0.25I_{30}$ B $0.33I_{30}$ C $0.75I_{30}$ D $0.87I_{30}$

Ans: B

Let I_0 be the intensity of the light.

$$I_{30} = I_0 \cos^2 \theta = I_0 \cos^2 30^\circ = 0.75I_0$$

$$I_{60} = I_0 \cos^2 60^\circ = 0.25I_0$$

Comparing, $I_{60} = 0.33I_{30}$

